

Titus Loftin

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EDUCATION

The University of Texas at Austin	Bachelor of Science, Chemical Engineering, Honors Minors and Certificates: Computational Science and Engineering, Programming and Computation, and Statistics and Data Science Overall GPA: 3.80	May 2027
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EXPERIENCE

University of Texas at Austin - Undergraduate Tutor	Aug. 2025 - Present
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- Tutored for ChE 210, covering over 100 undergraduates
- Facilitated group and one-on-one sessions to strengthen conceptual understanding and proficiency

Soft Tissue Biomechanics Lab - Undergraduate Researcher	Sept. 2024 - Present
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- Developed a method of identifying heterogeneous soft materials by collecting 8 different types of strain data from 120 different materials, and using neural ODE fields for mechanical characterization
- Created an in-vitro setup to run mechanical tests on blood clots in the middle cerebral artery that measures pressure and flow conditions up to an accuracy of 0.1%

Hill Country bible Church - Facilities Technical Intern	June 2020 - August 2024
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- Designed and built portable baptism tubs for use in the church, reducing potential costs by ~\$2K if bought from a third party
- Decreased energy use by 15% installing lighting fixtures throughout the facilities
- Installed a more efficient pond irrigation system to optimize water use by 8%

PROJECTS

Clarity

- Created a custom chemical process design and simulation platform with a Rust-based engine. Built a front-end interface with Tauri for constructing and running process simulations.

ReactorPy/ThermSolve

- Developed two Python Libraries that grant the ability to analyze and solve complex chemical engineering systems.

Excel Optimizer Plugin

- Designed a plugin for Excel for different chemical engineering calculations, specifically for optimization problems.

SKILLS

Programming Languages: Python, MATLAB, Julia, C++, Rust

Design Tools: SolidWorks, Fusion360, LabVIEW, OpenFOAM, Aspen Plus

LEADERSHIP EXPERIENCE AND ACTIVITIES

Chemical Engineering Undergraduate Advisory Board - Director and Founder	Fall 2024 - Present
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- Founded a student organization to improve the CHE department. Recruited ~25% of the Chemical Engineering student body during the 2024-2025 school year.

Tau Beta Pi - Vice President Corporate Relations, Fundraising Coordinator	Spring 2025 - Present
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- Managed relationships with companies like Blue Origin and Lockheed Martin, brought in corporate sponsors such as CDM Smith. Raised over \$2000 in the Fall 2025 semester.